

हिन्दुस्तान पेट्रोलियम कॉर्पोरेशन लिमिटेड

Hindustan Petroleum Corporation Limited

(A Maharatna Company)

Participant ID	
Participant Name	
Test Center Name	
Test Date	18/08/2024
Test Time	9:00 AM - 11:30 AM
Subject	Chemical Engineer

Section : English Language

Q.1 Select the most appropriate ANTONYM of the underlined word in the given sentence.

The transient nature of the artist's latest exhibition, which was only available for a brief period, created a sense of urgency among art enthusiasts.

- Ans
1. Temporary
2. Ephemeral
3. Fleeting
4. Permanent

Question ID : 630680941483

Option 1 ID : 6306803690474

Option 2 ID : 6306803690475

Option 3 ID : 6306803690477

Option 4 ID : 6306803690476

Status : Answered

Chosen Option : 4

Q.2 Select the INCORRECTLY spelt word.

- Ans
1. Occasion
2. Licenzing
3. Prejudice
4. Reference

Question ID : 630680461982

Option 1 ID : 6306801804144

Option 2 ID : 6306801804145

Option 3 ID : 6306801804142

Option 4 ID : 6306801804143

Status : Answered

Chosen Option : 2

Q.3 Select the most appropriate meaning of the underlined idiom.

While working on the project, the team had their backs to the wall.

- Ans
1. Was met with a lot of opposition
2. Had very serious problems
3. Was cut off from the society
4. Had a very serious deadline

Question ID : 630680987567

Option 1 ID : 6306803874266

Option 2 ID : 6306803874265

Option 3 ID : 6306803874263

Option 4 ID : 6306803874264

Status : Answered

Chosen Option : 1

Q.4 Select the most appropriate option that can substitute the underlined words in the given sentence.

In the new marketing campaign, the innovative product was the thing that was so beautiful that it attracted a lot of attention.

- Ans ☒ 1. contentious
☒ 2. exotic
☒ 3. a cynosure
☒ 4. ceremonious

Question ID : 630680987515
Option 1 ID : 6306803874058
Option 2 ID : 6306803874056
Option 3 ID : 6306803874057
Option 4 ID : 6306803874055
Status : Marked For Review
Chosen Option : 3

Q.5 Select the most appropriate option to fill in the blanks.

The _____ student in the class managed to solve the _____ maths problem within minutes, impressing the _____ teacher.

- Ans ☒ 1. intelligent; often; Indian
☒ 2. loudest; extremely; old
☒ 3. male; fairly; great
☒ 4. brightest; complex; experienced

Question ID : 630680940907
Option 1 ID : 6306803688133
Option 2 ID : 6306803688134
Option 3 ID : 6306803688132
Option 4 ID : 6306803688131
Status : Answered
Chosen Option : 4

Comprehension:

Read the given passage and answer the questions that follow.

The UK is one of the world's richest countries, but people are finding it increasingly difficult to pay energy bills. Electricity and gas bills could rise fourfold by the winter. Even people on good salaries are feeling the pinch. Lower-paid workers and those who have retired may soon be faced with a stark choice of heating or eating.

Local governments have devised a makeshift solution. They are converting communal areas like libraries into spaces where people can go to stay warm for free. These are called 'heat banks'. Money-saving expert Martin Lewis said he is dumbfounded by the idea of warm banks in a supposedly wealthy country. The UK already has hundreds of food banks for those who struggle to afford to eat. Britons are already taking drastic measures to avoid skyrocketing energy costs. One woman has disconnected her refrigerator and is storing perishable goods in cold water. A man has turned off his electric kettle and is drinking water instead of hot coffee.

Last winter, an older lady used her free bus pass to sit on a warm bus all day instead of sitting at home in the cold. Many people are furious with the government and fossil fuel companies. Giant oil companies are making record profits, and the government is reluctant to implement a one-off 'windfall tax'. The UK's National Energy Action group said that each year, around 11,400 deaths are caused by cold weather. They predict this figure will rise further this year.

SubQuestion No : 6

Q.6 What does the National Energy Action group predict will happen this year due to the energy crisis?

- Ans ☒ 1. A rise in the number of deaths caused by cold weather
☒ 2. An increase in the number of food banks
☒ 3. Implementation of a windfall tax by the government
☒ 4. A significant decrease in energy consumption

Question ID : 630680982494
Option 1 ID : 6306803853881
Option 2 ID : 6306803853882
Option 3 ID : 6306803853883
Option 4 ID : 6306803853880
Status : Answered
Chosen Option : 1

Comprehension:

Read the given passage and answer the questions that follow.

The UK is one of the world's richest countries, but people are finding it increasingly difficult to pay energy bills. Electricity and gas bills could rise fourfold by the winter. Even people on good salaries are feeling the pinch. Lower-paid workers and those who have retired may soon be faced with a stark choice of heating or eating.

Local governments have devised a makeshift solution. They are converting communal areas like libraries into spaces where people can go to stay warm for free. These are called 'heat banks'. Money-saving expert Martin Lewis said he is dumbfounded by the idea of warm banks in a supposedly wealthy country. The UK already has hundreds of food banks for those who struggle to afford to eat. Britons are already taking drastic measures to avoid skyrocketing energy costs. One woman has disconnected her refrigerator and is storing perishable goods in cold water. A man has turned off his electric kettle and is drinking water instead of hot coffee.

Last winter, an older lady used her free bus pass to sit on a warm bus all day instead of sitting at home in the cold. Many people are furious with the government and fossil fuel companies. Giant oil companies are making record profits, and the government is reluctant to implement a one-off 'windfall tax'. The UK's National Energy Action group said that each year, around 11,400 deaths are caused by cold weather. They predict this figure will rise further this year.

SubQuestion No : 7

Q.7 Which of the following titles best fits the given passage?

Ans ☒ 1. The Impact of Cold Weather on Public Health

☐ 2. The Expensive Heat Banks in the UK

☐ 3. The UK Government's Energy Policies

☐ 4. The Rising Profits of Fossil Fuel Companies

Question ID : 630680982492

Option 1 ID : 6306803853874

Option 2 ID : 6306803853873

Option 3 ID : 6306803853875

Option 4 ID : 6306803853872

Status : Answered

Chosen Option : 1

Comprehension:

Read the given passage and answer the questions that follow.

The UK is one of the world's richest countries, but people are finding it increasingly difficult to pay energy bills. Electricity and gas bills could rise fourfold by the winter. Even people on good salaries are feeling the pinch. Lower-paid workers and those who have retired may soon be faced with a stark choice of heating or eating.

Local governments have devised a makeshift solution. They are converting communal areas like libraries into spaces where people can go to stay warm for free. These are called 'heat banks'. Money-saving expert Martin Lewis said he is dumbfounded by the idea of warm banks in a supposedly wealthy country. The UK already has hundreds of food banks for those who struggle to afford to eat. Britons are already taking drastic measures to avoid skyrocketing energy costs. One woman has disconnected her refrigerator and is storing perishable goods in cold water. A man has turned off his electric kettle and is drinking water instead of hot coffee.

Last winter, an older lady used her free bus pass to sit on a warm bus all day instead of sitting at home in the cold. Many people are furious with the government and fossil fuel companies. Giant oil companies are making record profits, and the government is reluctant to implement a one-off 'windfall tax'. The UK's National Energy Action group said that each year, around 11,400 deaths are caused by cold weather. They predict this figure will rise further this year.

SubQuestion No : 8

Q.8 What solution have local governments in the UK devised to help people stay warm?

Ans ☐ 1. Providing subsidies for energy bills

☒ 2. Converting communal areas into heat banks

☐ 3. Distributing free blankets

☐ 4. Offering free electric heaters

Question ID : 630680982493

Option 1 ID : 6306803853877

Option 2 ID : 6306803853878

Option 3 ID : 6306803853876

Option 4 ID : 6306803853879

Status : Answered

Chosen Option : 2

Comprehension:

Read the given passage and answer the questions that follow.

The UK is one of the world's richest countries, but people are finding it increasingly difficult to pay energy bills. Electricity and gas bills could rise fourfold by the winter. Even people on good salaries are feeling the pinch. Lower-paid workers and those who have retired may soon be faced with a stark choice of heating or eating.

Local governments have devised a makeshift solution. They are converting communal areas like libraries into spaces where people can go to stay warm for free. These are called 'heat banks'. Money-saving expert Martin Lewis said he is dumbfounded by the idea of warm banks in a supposedly wealthy country. The UK already has hundreds of food banks for those who struggle to afford to eat. Britons are already taking drastic measures to avoid skyrocketing energy costs. One woman has disconnected her refrigerator and is storing perishable goods in cold water. A man has turned off his electric kettle and is drinking water instead of hot coffee.

Last winter, an older lady used her free bus pass to sit on a warm bus all day instead of sitting at home in the cold. Many people are furious with the government and fossil fuel companies. Giant oil companies are making record profits, and the government is reluctant to implement a one-off 'windfall tax'. The UK's National Energy Action group said that each year, around 11,400 deaths are caused by cold weather. They predict this figure will rise further this year.

SubQuestion No : 9

Q.9 Based on the given passage, which of the following can be inferred about the public's sentiment towards the government and fossil fuel companies?

- Ans
- ☒ 1. The public feels grateful for the measures taken by the government.
 - ☒ 2. The public is largely supportive of the government's energy policies.
 - ☒ 3. The public is angry with the government and fossil fuel companies.
 - ☒ 4. The public is indifferent to the profits made by fossil fuel companies.

Question ID : 630680982495

Option 1 ID : 6306803853887

Option 2 ID : 6306803853884

Option 3 ID : 6306803853886

Option 4 ID : 6306803853885

Status : Answered

Chosen Option : 3

Comprehension:

Read the given passage and answer the questions that follow.

The UK is one of the world's richest countries, but people are finding it increasingly difficult to pay energy bills. Electricity and gas bills could rise fourfold by the winter. Even people on good salaries are feeling the pinch. Lower-paid workers and those who have retired may soon be faced with a stark choice of heating or eating.

Local governments have devised a makeshift solution. They are converting communal areas like libraries into spaces where people can go to stay warm for free. These are called 'heat banks'. Money-saving expert Martin Lewis said he is dumbfounded by the idea of warm banks in a supposedly wealthy country. The UK already has hundreds of food banks for those who struggle to afford to eat. Britons are already taking drastic measures to avoid skyrocketing energy costs. One woman has disconnected her refrigerator and is storing perishable goods in cold water. A man has turned off his electric kettle and is drinking water instead of hot coffee.

Last winter, an older lady used her free bus pass to sit on a warm bus all day instead of sitting at home in the cold. Many people are furious with the government and fossil fuel companies. Giant oil companies are making record profits, and the government is reluctant to implement a one-off 'windfall tax'. The UK's National Energy Action group said that each year, around 11,400 deaths are caused by cold weather. They predict this figure will rise further this year.

SubQuestion No : 10

Q.10 Which of the following words is a synonym for the given word as used in the given passage?

Dumbfounded

- Ans
- ☒ 1. Indifferent
 - ☒ 2. Delighted
 - ☒ 3. Astonished
 - ☒ 4. Confused

Question ID : 630680982496

Option 1 ID : 6306803889764

Option 2 ID : 6306803853888

Option 3 ID : 6306803853890

Option 4 ID : 6306803853889

Status : Answered

Chosen Option : 4

Q.11 Select the most appropriate option to fill in the blank.

The professor's lectures are always _____ than those of his colleagues.

Ans ☒ 1. most engaging
☒ 2. the most engaging
☒ 3. more engaging
☒ 4. engaging

Question ID : 630680940888
Option 1 ID : 6306803688057
Option 2 ID : 6306803688058
Option 3 ID : 6306803688056
Option 4 ID : 6306803688055
Status : Answered
Chosen Option : 3

Q.12 Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph.

A. She decided to move to Delhi to her father's place in search of work with only ₹500 in her pocket.
B. Born in the small village of Daulatpur in Uttar Pradesh, Krishna Yadav never attended school.
C. "Later, he tried to start a vehicle trading business, but we incurred huge losses. It became increasingly difficult for us to survive. We couldn't afford to eat vegetables anymore and survived on namak roti (salt and flatbread) for days," says Krishna.
D. After her marriage, she moved to Bulandshahr, where she faced a dire situation when her husband lost his job as a traffic police officer.

Ans ☒ 1. BDCA
☒ 2. ADBC
☒ 3. CADB
☒ 4. DACB

Question ID : 630680983987
Option 1 ID : 6306803860018
Option 2 ID : 6306803860020
Option 3 ID : 6306803860017
Option 4 ID : 6306803860019
Status : Answered
Chosen Option : 1

Q.13 Select the most appropriate meaning of the given idiom.

Spill the beans

Ans ☒ 1. To waste resources
☒ 2. To cook a meal
☒ 3. To make a mistake
☒ 4. To reveal a secret

Question ID : 630680941311
Option 1 ID : 6306803689795
Option 2 ID : 6306803689794
Option 3 ID : 6306803689797
Option 4 ID : 6306803689796
Status : Answered
Chosen Option : 4

Q.14 Select the most appropriate option to fill in the blank.

_____ Philippines is an archipelago comprising over 7,000 islands.

Ans ☒ 1. An
☒ 2. The
☒ 3. A
☒ 4. It

Question ID : 630680940885
Option 1 ID : 6306803688036
Option 2 ID : 6306803688037
Option 3 ID : 6306803688035
Option 4 ID : 6306803688038
Status : Answered
Chosen Option : 2

Q.15 Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph.

- A. It is expected the installation, described by Mr. Furman as a "swirling, wild, exciting colourful piece of art", will be seen by about 2.1 million people who go past the wall each year.
B. "It's next to the actual bridge over the River Thames and I was really inspired by the history of the site," he said.
C. The 57-m (187-ft) artwork outside London Bridge Station has been completed.
D. A new mosaic near London Bridge, made from 2,50,000 small glass pieces placed by hand, celebrates the diversity of the local area, its designer says.

Ans ☒ 1. ADBC

☒ 2. DCAB

☒ 3. CADB

☒ 4. BADC

Question ID : 6306801026933

Option 1 ID : 6306804030329

Option 2 ID : 6306804030326

Option 3 ID : 6306804030328

Option 4 ID : 6306804030327

Status : Not Answered

Chosen Option : --

Q.16 Select the most appropriate option to fill in the blank.

Out of all teachers, Mrs. Sampath is _____ by the students.

Ans ☒ 1. admired

☒ 2. admirable

☒ 3. more admired

☒ 4. most admired

Question ID : 6306801026938

Option 1 ID : 6306804030347

Option 2 ID : 6306804030349

Option 3 ID : 6306804030346

Option 4 ID : 6306804030348

Status : Answered

Chosen Option : 4

Q.17 Select the most appropriate ANTONYM of the underlined word in the given sentence.

The atmosphere in the meeting was cordial, promoting open and friendly discussions.

Ans ☒ 1. Warm

☒ 2. Amiable

☒ 3. Congenial

☒ 4. Hostile

Question ID : 6306801026899

Option 1 ID : 6306804030228

Option 2 ID : 6306804030227

Option 3 ID : 6306804030229

Option 4 ID : 6306804030226

Status : Not Answered

Chosen Option : --

Section : Quantitative Aptitude

Q.1 The area (in cm^2) of an equilateral triangle of side 100 cm is:

Ans ☒ 1. $2497\sqrt{3}$

☒ 2. $2500\sqrt{3}$

☒ 3. $2505\sqrt{3}$

☒ 4. $2503\sqrt{3}$

Question ID : 630680841406

Option 1 ID : 6306803296180

Option 2 ID : 6306803296177

Option 3 ID : 6306803296178

Option 4 ID : 6306803296179

Status : Answered

Chosen Option : 2

Q.2 What will be difference in population 3 years ago and 2 years ago of a town, whose current population is 470000 and which is increasing at a rate of 25% every year?

- Ans
- ☒ 1. 59410
 - ☒ 2. 59610
 - ☒ 3. 60510
 - ☒ 4. 60160

Question ID : 630680998298
Option 1 ID : 6306803917097
Option 2 ID : 6306803917096
Option 3 ID : 6306803917098
Option 4 ID : 6306803917095
Status : Not Answered
Chosen Option : --

Q.3 A man spends 20% of his monthly salary (in ₹) on the rent of his house. If every month he also spends ₹232 on his conveyance and ₹9307 on his grocery and saves the remaining ₹553, his monthly salary (in ₹) is:

- Ans
- ☒ 1. 12615
 - ☒ 2. 12637
 - ☒ 3. 12600
 - ☒ 4. 12555

Question ID : 6306801026872
Option 1 ID : 6306804030134
Option 2 ID : 6306804030136
Option 3 ID : 6306804030135
Option 4 ID : 6306804030137
Status : Not Answered
Chosen Option : --

Q.4 State the nature of the roots of the following quadratic equation.
 $5x^2 + 2 + 4x = 0$

- Ans
- ☒ 1. No real roots
 - ☒ 2. Roots are real and distinct
 - ☒ 3. Both roots are negative
 - ☒ 4. Roots are equal

Question ID : 630680180174
Option 1 ID : 630680698441
Option 2 ID : 630680698442
Option 3 ID : 630680698443
Option 4 ID : 630680698440
Status : Answered
Chosen Option : 3

Q.5 A shopkeeper marks an article at ₹x and offers a discount of 40% on it. He sells it for ₹429 after charging VAT of 30% on the discounted price. What is the value of ₹ x?

- Ans
- ☒ 1. ₹550
 - ☒ 2. ₹400
 - ☒ 3. ₹600
 - ☒ 4. ₹500

Question ID : 6306801026882
Option 1 ID : 6306804030522
Option 2 ID : 6306804030525
Option 3 ID : 6306804030523
Option 4 ID : 6306804030524
Status : Not Answered
Chosen Option : --

Q.6 The sides of a triangle are 39 cm, 15 cm, and 36 cm. What is its area? (in cm^2)

- Ans
- ☒ 1. 258
 - ☒ 2. 272
 - ☒ 3. 298
 - ☒ 4. 270

Question ID : 6306801026876
Option 1 ID : 6306804030152
Option 2 ID : 6306804030151
Option 3 ID : 6306804030153
Option 4 ID : 6306804030150
Status : Answered
Chosen Option : 3

Q.7 If $3.8 : 13.7 :: 19 : x$, find the value of x .

- Ans
- ☒ 1. 70.2
 - ☒ 2. 68.5
 - ☒ 3. 70.1
 - ☒ 4. 69.3

Question ID : 6306801026893
Option 1 ID : 6306804030532
Option 2 ID : 6306804030530
Option 3 ID : 6306804030533
Option 4 ID : 6306804030531
Status : Not Answered
Chosen Option : --

Q.8 Find the time taken by a train of length 384 m running at a speed of 125 km/h to cross another train of length 456 m running at a speed of 119 km/h in the same direction.

- Ans
- ☒ 1. 1400 seconds
 - ☒ 2. 846 seconds
 - ☒ 3. 140 seconds
 - ☒ 4. 504 seconds

Question ID : 6306801026849
Option 1 ID : 6306804030059
Option 2 ID : 6306804030061
Option 3 ID : 6306804030060
Option 4 ID : 6306804030058
Status : Not Answered
Chosen Option : --

Q.9 If $99 : y :: y : 176$, find the positive value of y .

- Ans
- ☒ 1. 129
 - ☒ 2. 135
 - ☒ 3. 132
 - ☒ 4. 125

Question ID : 630680789467
Option 1 ID : 6306803093650
Option 2 ID : 6306803093649
Option 3 ID : 6306803093648
Option 4 ID : 6306803093651
Status : Not Answered
Chosen Option : --

Q.10 Chetan invested ₹1500 at 5% per annum simple interest in a bank. What amount (in ₹) will he get after 6 years?

- Ans
- ☒ 1. 1950
 - ☒ 2. 450
 - ☒ 3. 550
 - ☒ 4. 1850

Question ID : 630680666015
Option 1 ID : 6306802607620
Option 2 ID : 6306802607621
Option 3 ID : 6306802607623
Option 4 ID : 6306802607622
Status : Answered
Chosen Option : 1

Q.11 The area (in cm^2) of an equilateral triangle of side 28 cm is:

- Ans
- ☒ 1. $199\sqrt{3}$
 - ☒ 2. $191\sqrt{3}$
 - ☒ 3. $196\sqrt{3}$
 - ☒ 4. $192\sqrt{3}$

Question ID : 630680841408
Option 1 ID : 6306803296187
Option 2 ID : 6306803296188
Option 3 ID : 6306803296185
Option 4 ID : 6306803296186
Status : Answered
Chosen Option : 3

Q.12 The wages of 5 masons for a 2-day week are ₹1650. What are the one-day wage (in ₹) of one worker?

- Ans
- ☐ 1. 330
 - ☐ 2. 166
 - ☐ 3. 331
 - ☒ 4. 165

Question ID : 630680660510
Option 1 ID : 6306802585983
Option 2 ID : 6306802585984
Option 3 ID : 6306802585985
Option 4 ID : 6306802585982
Status : Answered
Chosen Option : 4

Q.13 The population of a village was 110000. It increased by 5% in the first year and increased by 25% in the second year. Its population after two years is _____.

- Ans
- ☐ 1. 115500
 - ☒ 2. 144375
 - ☐ 3. 143000
 - ☐ 4. 137500

Question ID : 630680591513
Option 1 ID : 6306802314520
Option 2 ID : 6306802314519
Option 3 ID : 6306802314522
Option 4 ID : 6306802314521
Status : Answered
Chosen Option : 2

Q.14 The average of 11 results is 61. If the average of the first 6 results is 64 and that of the last 6 is 62, find the 6th result.

- Ans
- ☐ 1. 84
 - ☐ 2. 87
 - ☒ 3. 85
 - ☐ 4. 86

Question ID : 6306801026860
Option 1 ID : 6306804030513
Option 2 ID : 6306804030512
Option 3 ID : 6306804030510
Option 4 ID : 6306804030511
Status : Not Answered
Chosen Option : --

Q.15 Evaluate

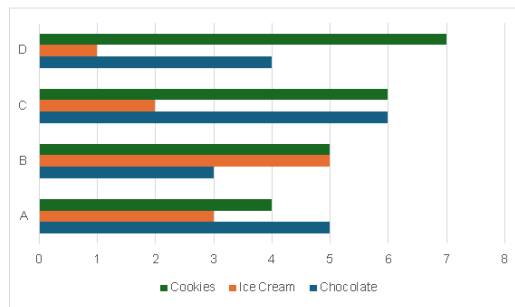
$$\sqrt{54 + \sqrt{\left(91 + \sqrt{\left(73 + \sqrt{\left(58 + \sqrt{(36)}\right)}\right)}\right)}}$$

- Ans
- ☐ 1. 4
 - ☒ 2. 8
 - ☐ 3. 11
 - ☐ 4. 6

Question ID : 630680704899
Option 1 ID : 6306802761285
Option 2 ID : 6306802761282
Option 3 ID : 6306802761284
Option 4 ID : 6306802761283
Status : Answered
Chosen Option : 2

Q.16 Study the following graph carefully and answer the question given below it.

Number of Items (in thousands) i.e. Cookies, Ice cream and Chocolate production by different companies.



What is the difference between total production of Company B and D?

Ans ☒ 1. 1000

☒ 2. 10

☒ 3. 100

☒ 4. 1

Question ID : 630680998387
Option 1 ID : 6306803917450
Option 2 ID : 6306803917453
Option 3 ID : 6306803917452
Option 4 ID : 6306803917451
Status : Answered
Chosen Option : 1

Q.17 The average weight of a family of five members whose weights are 40 kg, 49 kg, 57 kg, 63 kg and 36 kg is:

Ans ☒ 1. 49

☒ 2. 50

☒ 3. 51

☒ 4. 48

Question ID : 630680657011
Option 1 ID : 6306802571986
Option 2 ID : 6306802571987
Option 3 ID : 6306802571989
Option 4 ID : 6306802571988
Status : Answered
Chosen Option : 1

Q.18 When x is added to each of 39, 27, 34 and 36, then the numbers so obtained, in this order, are in proportion. Then, if $3x : y :: y : (7x-9)$, and $y > 0$, what is the value of y ?

Ans ☒ 1. 181

☒ 2. 162

☒ 3. 169

☒ 4. 147

Question ID : 6306801026904
Option 1 ID : 6306804030241
Option 2 ID : 6306804030238
Option 3 ID : 6306804030239
Option 4 ID : 6306804030240
Status : Not Answered
Chosen Option : --

Q.19 A company earns a profit (in ₹) that is distributed among the company's three partners in the ratio of 2 : 19 : 13. If the difference between the smallest and the largest shares is ₹14866, the total profit (in ₹) of the company is:

Ans ☒ 1. 29758

☒ 2. 29732

☒ 3. 29692

☒ 4. 29792

Question ID : 630680789364
Option 1 ID : 6306803093237
Option 2 ID : 6306803093236
Option 3 ID : 6306803093239
Option 4 ID : 6306803093238
Status : Not Answered
Chosen Option : --

Q.20 A, B, C alone can do a piece of work in 11, 66 and 88 days respectively. They all started the work together, but A left after 4 days. In how many days, was the remaining work completed?

- Ans
- ☐ 1. 19
 - ☐ 2. 21
 - ☒ 3. 20
 - ☐ 4. 22

Question ID : 6306801026862
Option 1 ID : 6306804030101
Option 2 ID : 6306804030099
Option 3 ID : 6306804030098
Option 4 ID : 6306804030100
Status : Not Answered
Chosen Option : --

Q.21 Evaluate: $38 - 8 \div 16 \times 4$

- Ans
- ☐ 1. 35
 - ☒ 2. 36
 - ☐ 3. 34
 - ☐ 4. 39

Question ID : 6306801026866
Option 1 ID : 6306804030111
Option 2 ID : 6306804030110
Option 3 ID : 6306804030112
Option 4 ID : 6306804030113
Status : Answered
Chosen Option : 2

Q.22 The value of the discriminant of the quadratic equation $13x^2 - 15x - 16 = 0$ is:

- Ans
- ☒ 1. 1057
 - ☐ 2. 1042
 - ☐ 3. 1071
 - ☐ 4. 1047

Question ID : 630680841531
Option 1 ID : 6306803296677
Option 2 ID : 6306803296679
Option 3 ID : 6306803296680
Option 4 ID : 6306803296678
Status : Answered
Chosen Option : 2

Q.23 The value of $9^3 - 2^2 + \left(\frac{10}{2}\right)^2 - 1 + 5 \times 0 =$ _____

- Ans
- ☐ 1. 752
 - ☐ 2. 758
 - ☐ 3. 747
 - ☒ 4. 749

Question ID : 630680676726
Option 1 ID : 6306802650352
Option 2 ID : 6306802650351
Option 3 ID : 6306802650350
Option 4 ID : 6306802650349
Status : Answered
Chosen Option : 4

Q.24 A man sells two articles for ₹20000 each. He gets a profit of 70% on one article and loses 70% on the other article. What is his overall profit or loss (to the nearest integer) in this transaction?

- Ans
- ☐ 1. ₹38531 loss
 - ☒ 2. ₹38431 loss
 - ☐ 3. ₹38531 profit
 - ☐ 4. ₹38431 profit

Question ID : 630680790043
Option 1 ID : 6306803095954
Option 2 ID : 6306803095952
Option 3 ID : 6306803095955
Option 4 ID : 6306803095953
Status : Not Answered
Chosen Option : --

Q.25

Gopal invested ₹1500 at 5% per annum simple interest in a bank. What amount (in ₹) will he get after 3 years?

Ans

✗ 1. 225

✗ 2. 1625

✓ 3. 1725

✗ 4. 325

Question ID : 6306801026863

Option 1 ID : 6306804030103

Option 2 ID : 6306804030104

Option 3 ID : 6306804030102

Option 4 ID : 6306804030105

Status : Answered

Chosen Option : 2

Q.26

The population of a village was 110000. It increased by 10% in the first year and increased by 20% in the second year. Its population after two years is _____.

Ans

✗ 1. 132000

✗ 2. 121000

✗ 3. 143000

✓ 4. 145200

Question ID : 6306801026857

Option 1 ID : 6306804030084

Option 2 ID : 6306804030083

Option 3 ID : 6306804030085

Option 4 ID : 6306804030082

Status : Answered

Chosen Option : 1

Q.27

Robin has a few coins of 1 rupee, 50 paise and 25 paise in the ratio $\frac{1}{4} : \frac{1}{2} : \frac{1}{2}$. If the number of 50 paise coins is 200, then

Ans

✗ 1. ₹125

✗ 2. ₹200

✓ 3. ₹250

✗ 4. ₹100

Question ID : 630680998313

Option 1 ID : 6306803917155

Option 2 ID : 6306803917156

Option 3 ID : 6306803917158

Option 4 ID : 6306803917157

Status : Answered

Chosen Option : 3

Q.28

From a taxi stand, two cabs start at a speed of 47 km/h at an interval of 47 minutes, both cabs travelling in the same direction. A man coming in the opposite direction towards the taxi stand meets the cabs at an interval of 10 minutes. Find the speed of the man.

Ans

✓ 1. 173.9

✗ 2. 166.5

✗ 3. 183.4

✗ 4. 167.2

Question ID : 630680998313

Option 1 ID : 6306803917155

Option 2 ID : 6306803917156

Option 3 ID : 6306803917158

Option 4 ID : 6306803917157

Status : Answered

Chosen Option : 3

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12/40

Q.29 Bablu walks to his farm house 3 km away from home. In the morning he covers the distance in 1 hour whereas, while returning home in the evening, he takes 60 more minutes to cover the same distance. Find his average speed (in kmph) during the two-way journey.

- Ans ☒ 1. 2.5
☒ 2. 2
☒ 3. 3
☒ 4. 3.5

Question ID : 630680998309
Option 1 ID : 6306803917140
Option 2 ID : 6306803917139
Option 3 ID : 6306803917141
Option 4 ID : 6306803917142
Status : Answered
Chosen Option : 4

Q.30 The length of a rectangular fenced area is 15 m more than its breadth. Its area is 31096 m². Its breadth (in m) is:

- Ans ☒ 1. 149
☒ 2. 170
☒ 3. 169
☒ 4. 151

Question ID : 6306801026880
Option 1 ID : 6306804030521
Option 2 ID : 6306804030520
Option 3 ID : 6306804030518
Option 4 ID : 6306804030519
Status : Answered
Chosen Option : 3

Q.31 The product of two consecutive positive integers is 952. If the smaller of these two integers is represented by x, which of the options below will correspond to the equation for finding out the value of x?

- Ans ☒ 1. $x^2 - x - 952 = 0$
☒ 2. $x^2 - x + 952 = 0$
☒ 3. $x^2 + x - 952 = 0$
☒ 4. $x^2 + x + 952 = 0$

Question ID : 630680841794
Option 1 ID : 6306803297731
Option 2 ID : 6306803297732
Option 3 ID : 6306803297729
Option 4 ID : 6306803297730
Status : Not Answered
Chosen Option : --

Q.32 A person drove at 79 kmph for 3.5 hours, then increased his speed by 18 kmph, and reached the destination in another hour. Find the average speed (in kmph) of the person during the entire journey.

- Ans ☒ 1. 83
☒ 2. 87
☒ 3. 79
☒ 4. 78

Question ID : 630680894223
Option 1 ID : 6306803502909
Option 2 ID : 6306803502911
Option 3 ID : 6306803502910
Option 4 ID : 6306803502912
Status : Not Answered
Chosen Option : --

Q
3
3
A
n
s

The HCF and the LCM of two numbers are 36 and 180, respectively. If one of the numbers is $\frac{5}{4}$ times of the other, what is the

✖ 1. 104

✔ 2. 123

✔ 3. 90

✖ 4. 96

Question ID : 6306807
Option 1 ID : 6306803
Option 2 ID : 6306803
Option 3 ID : 6306803
Option 4 ID : 6306803
Status : Not Ans
Chosen Option : --

Q.34

The average of 11 results is 61. If the average of the first 6 results is 64 and that of the last 6 is 63, find the 6th result.

Ans

✔ 1. 91

✖ 2. 93

✖ 3. 90

✖ 4. 92

Question ID : 630680657673
Option 1 ID : 6306802574634
Option 2 ID : 6306802574636
Option 3 ID : 6306802574637
Option 4 ID : 6306802574635
Status : Not Answered
Chosen Option : --

Section : Intellectual Potential Test

Q.1

In a certain code language, 'HOPS' is coded as '2031' and 'PAOS' is coded as '0413'.What is the code for 'A' in the given code language?

Ans

✖ 1. 1

✔ 2. 4

✖ 3. 0

✖ 4. 3

Question ID : 6306801026884
Option 1 ID : 6306804030176
Option 2 ID : 6306804030175
Option 3 ID : 6306804030174
Option 4 ID : 6306804030177
Status : Answered
Chosen Option : 2

Q.2

Each of D, E, G, H, K, L and S were born on different days of the same week starting from Monday and ending on Sunday.
D was born on one of the days before Wednesday. Only two people were born between D and E. As many people were born after E as were born before G. H was born immediately after S. K was born on one of the days before L. Who was born on Thursday?

Ans

✖ 1. E

✖ 2. S

✖ 3. K

✔ 4. L

Question ID : 630680152958
Option 1 ID : 630680591773
Option 2 ID : 630680591775
Option 3 ID : 630680591774
Option 4 ID : 630680591772
Status : Not Answered
Chosen Option : --

Q.3 Seven people A, B, C, D, P, Q and R are sitting around a circular table, facing the centre. D is sitting to the immediate left of C. Q is sitting third to the right of P. Q is an immediate neighbour of neither D nor C. A is an immediate neighbour of Q but not of B. B is not an immediate neighbour of Q. Who is sitting fourth to the left of D?

Ans  1. P
 2. Q
 3. B
 4. R

Question ID : 630680341551
Option 1 ID : 6306801328638
Option 2 ID : 6306801328639
Option 3 ID : 6306801328637
Option 4 ID : 6306801328640
Status : Answered
Chosen Option : 3

Q.4 Refer to the following letter, number and symbol series and answer the question that follows. Counting to be done from left to right only.

(Left) & 2 R Y # G C & T 3 5 E * S % 7 D 4 @ 6 K (Right)

How many such symbols are there which are immediately preceded by a number and also immediately followed by a letter?

Ans  1. 2
 2. 3
 3. 0
 4. 1

Question ID : 630680908114
Option 1 ID : 6306803557532
Option 2 ID : 6306803557533
Option 3 ID : 6306803557530
Option 4 ID : 6306803557531
Status : Answered
Chosen Option : 4

Q.5 The following contains two pairs of words which are related to each other in a certain way. Three of the following four word-pairs are alike as these have the same relationship and thus form a group. Which word-pair is the one that does NOT belong to that group?
(The words must be considered as meaningful English words and must not be related to each other based on the number of letters/number of consonants/vowels in the word.)

Ans  1. Decline – Reduce
 2. Grab – Seize
 3. Hate – Like
 4. Defeat – Conquer

Question ID : 6306801026858
Option 1 ID : 6306804030086
Option 2 ID : 6306804030088
Option 3 ID : 6306804030089
Option 4 ID : 6306804030087
Status : Answered
Chosen Option : 1

Q.6 Sunil starts from Point A and drives 15 km towards West. He then takes a right turn, drives 4 km, turns right and drives 7 km. He takes a final right turn, drives 4 km and stops at Point P. How far (shortest distance) and towards which direction should he drive in order to reach Point A again? (All turns are 90-degree turns only unless otherwise specified.)

Ans  1. 6 km East
 2. 7 km East
 3. 5 km East
 4. 8 km East

Question ID : 630680574742
Option 1 ID : 6306802247773
Option 2 ID : 6306802247774
Option 3 ID : 6306802247772
Option 4 ID : 6306802247775
Status : Answered
Chosen Option : 4

Q.7 The position(s) of how many digits will remain unchanged if each of the digits in the number 7324689 is arranged in ascending order from left to right?

- Ans ☒ 1. One
☒ 2. Three
☒ 3. None
☒ 4. Two

Question ID : 630680373814
Option 1 ID : 6306801456527
Option 2 ID : 6306801456529
Option 3 ID : 6306801456530
Option 4 ID : 6306801456528
Status : Answered
Chosen Option : 2

Q.8 Seven boxes A, B, C, D, E, F and G are kept one over the other but not necessarily in the same order. Only B is kept above G. Only E is kept between G and C. Only F is kept below D. How many boxes are kept above A?

- Ans ☒ 1. Two
☒ 2. Four
☒ 3. Three
☒ 4. Five

Question ID : 630680908034
Option 1 ID : 6306803557206
Option 2 ID : 6306803557208
Option 3 ID : 6306803557207
Option 4 ID : 6306803557209
Status : Answered
Chosen Option : 2

Q.9 Refer to the following letter, number, symbol series and answer the question that follows.
(Left) 4 @ K E * D R Y # 3 C & 6 2 T % S G 7 5 & (Right)
If all the numbers are dropped from the series, which of the following will be 6th from the right?

- Ans ☒ 1. C
☒ 2. T
☒ 3. &
☒ 4. #

Question ID : 630680328576
Option 1 ID : 6306801278275
Option 2 ID : 6306801278273
Option 3 ID : 6306801278274
Option 4 ID : 6306801278276
Status : Answered
Chosen Option : 3

Q.10 What should come in place of the question mark (?) in the given series based on the English alphabetical order?

JLN, KMO, LNP, MOQ, ?

- Ans ☒ 1. NRP
☒ 2. MPR
☒ 3. NPR
☒ 4. MRP

Question ID : 6306801026845
Option 1 ID : 6306804030495
Option 2 ID : 6306804030496
Option 3 ID : 6306804030494
Option 4 ID : 6306804030497
Status : Answered
Chosen Option : 3

Q.11 Select the number from among the given options that can replace the question mark (?) in the following series.

12, 17, 27, 42, 62, ?

- Ans
- ☐ 1. 89
 - ☐ 2. 85
 - ☒ 3. 87
 - ☐ 4. 83

Question ID : 630680411221
Option 1 ID : 6306801603643
Option 2 ID : 6306801603642
Option 3 ID : 6306801603640
Option 4 ID : 6306801603641
Status : Answered
Chosen Option : 3

Q.12 What should come in place of the question mark (?) in the given series?

33 62 91 ? 149 178

- Ans
- ☐ 1. 140
 - ☒ 2. 120
 - ☐ 3. 112
 - ☐ 4. 131

Question ID : 630680510725
Option 1 ID : 6306801996033
Option 2 ID : 6306801996031
Option 3 ID : 6306801996030
Option 4 ID : 6306801996032
Status : Answered
Chosen Option : 2

Q.13 A, B, C, D, E, and F are sitting around a circular table facing the centre. D is an immediate neighbour of both B and A. F is sitting second to the right of D. A is not an immediate neighbour of E. C is an immediate neighbour of both F and E. Who is an immediate neighbour of both B and C?

- Ans
- ☐ 1. F
 - ☐ 2. A
 - ☐ 3. C
 - ☒ 4. E

Question ID : 6306801026853
Option 1 ID : 6306804030071
Option 2 ID : 6306804030070
Option 3 ID : 6306804030072
Option 4 ID : 6306804030073
Status : Answered
Chosen Option : 4

Q.14 A statement is followed by two conclusions. Identify which of the given conclusions is/are true based on the given statement.

Statement: $A \geq C = B < L \geq K$

Conclusions:

I. $C < L$

II. $A \geq K$

- Ans
- ☒ 1. Only conclusion I is true
 - ☐ 2. Only conclusion II is true
 - ☐ 3. Both conclusions I and II are true
 - ☐ 4. Neither conclusion I nor II is true

Question ID : 6306801026842
Option 1 ID : 6306804030039
Option 2 ID : 6306804030038
Option 3 ID : 6306804030040
Option 4 ID : 6306804030041
Status : Answered
Chosen Option : 2

Q.15 Three of the following word pairs are alike as they have the same relationship and hence form a group. Which word pair does NOT belong to that group?
(The words must be considered as meaningful English words and must not be grouped based on the number of letters/number of consonants/vowels in the word.)

- Ans ☒ 1. Pernicious - Malevolent
☒ 2. Enigma - Solution
☒ 3. Mellifluous - Dulcet
☒ 4. Salubrious - Wholesome

Question ID : 630680697399
Option 1 ID : 6306802731682
Option 2 ID : 6306802731681
Option 3 ID : 6306802731683
Option 4 ID : 6306802731684
Status : Not Answered
Chosen Option : --

Q.16 In a certain code language,
A + B means 'A is the mother of B',
A - B means 'A is the brother of B',
A × B means 'A is the wife of B',
A % B means 'A is the father of B' and
A # B means 'A is the daughter of B'.
Based on the above, how is N related to Y if 'N + Z - Q # R - Y % V'?

- Ans ☒ 1. Sister's daughter
☒ 2. Husband's sister
☒ 3. Brother's wife
☒ 4. Mother

Question ID : 630680503106
Option 1 ID : 6306801966012
Option 2 ID : 6306801966009
Option 3 ID : 6306801966011
Option 4 ID : 6306801966010
Status : Answered
Chosen Option : 2

Q.17 If 1 is added to each of the digits in the number 4065718 and the digits of the new number thus formed are re-arranged in ascending order from left to right, then which digit will be positioned equidistant from both the extreme ends?

- Ans ☒ 1. 3
☒ 2. 6
☒ 3. 7
☒ 4. 5

Question ID : 630680264614
Option 1 ID : 6306801027960
Option 2 ID : 6306801027961
Option 3 ID : 6306801027962
Option 4 ID : 6306801027959
Status : Answered
Chosen Option : 2

Q.18 In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets.

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 - Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

88, 303
73, 288

- Ans ☒ 1. 118, 323
☒ 2. 109, 314
☒ 3. 93, 318
☒ 4. 125, 340

Question ID : 630680799942
Option 1 ID : 6306803134560
Option 2 ID : 6306803134557
Option 3 ID : 6306803134558
Option 4 ID : 6306803134559
Status : Answered
Chosen Option : 3

Q.19 A, B, C, D, E, and F are sitting around a circular table facing the centre. B sits second to the right of C. A sits second to the left of C. D is the immediate neighbour of A and B. E is an immediate neighbour of both A and C. Who is an immediate neighbour of both E and D?

- Ans
- ☒ 1. C
 - ☒ 2. A
 - ☒ 3. B
 - ☒ 4. F

Question ID : 630680520386
Option 1 ID : 6306802033970
Option 2 ID : 6306802033971
Option 3 ID : 6306802033972
Option 4 ID : 6306802033969
Status : Answered
Chosen Option : 2

Q.20 OJQM is related to PKRN in a certain way based on the English alphabetical order. In the same way, RMTP is related to SNUQ. To which of the following is TOVR related following the same logic?

- Ans
- ☒ 1. PUSW
 - ☒ 2. UPWS
 - ☒ 3. UPSW
 - ☒ 4. PUWS

Question ID : 630680444386
Option 1 ID : 6306801734823
Option 2 ID : 6306801734820
Option 3 ID : 6306801734821
Option 4 ID : 6306801734822
Status : Answered
Chosen Option : 2

Q.21 15 is related to 217 following a certain logic. Following the same logic, 11 is related to 113. To which of the following is 8 related, following the same logic?

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

- Ans
- ☒ 1. 56
 - ☒ 2. 72
 - ☒ 3. 83
 - ☒ 4. 64

Question ID : 6306801026870
Option 1 ID : 6306804030126
Option 2 ID : 6306804030128
Option 3 ID : 6306804030129
Option 4 ID : 6306804030127
Status : Not Answered
Chosen Option : --

Q.22 What should come in place of the question mark (?) in the given series based on the English alphabetical order?

LOK, NQM, PSO, RUQ, ?

- Ans
- ☒ 1. SWT
 - ☒ 2. STW
 - ☒ 3. TSW
 - ☒ 4. TWS

Question ID : 630680426528
Option 1 ID : 6306801664201
Option 2 ID : 6306801664200
Option 3 ID : 6306801664199
Option 4 ID : 6306801664198
Status : Answered
Chosen Option : 4

Q.23 A statement is followed by two conclusions. Identify which of the given conclusion(s) is/are true based on the given statement.

Statement: $R > K = B \leq L < M > P$

Conclusions:

I. $R < B$

II. $M \leq K$

- Ans ☒ 1. Only conclusion I is true
- ☒ 2. Neither conclusion I nor II is true
- ☒ 3. Both conclusion I and II are true
- ☒ 4. Only conclusion II is true

Question ID : 630680418989
Option 1 ID : 6306801634670
Option 2 ID : 6306801634667
Option 3 ID : 6306801634668
Option 4 ID : 6306801634669
Status : Answered
Chosen Option : 1

Q.24 In the following number-pairs, the second number is obtained by performing certain mathematical operations on the first number. Which numbers should replace X and Y so that the pattern followed by the two numbers on the left side of :: is the same as that on the right side of :: ?

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 12 – Operations on 12 such as adding/subtracting/multiplying etc. to 12 can be performed. Breaking down 12 into 1 and 2 and then performing mathematical operations on 1 and 2 is not allowed.)

X : 118 :: 12 : Y

- Ans ☒ 1. X = 10, Y = 144
- ☒ 2. X = 11, Y = 141
- ☒ 3. X = 13, Y = 144
- ☒ 4. X = 11, Y = 147

Question ID : 6306801026838
Option 1 ID : 6306804030024
Option 2 ID : 6306804030025
Option 3 ID : 6306804030022
Option 4 ID : 6306804030023
Status : Answered
Chosen Option : 2

Q.25 A is the sister of B. B is the father of C. C is the sister of D. D is the husband of E. How is A related to E?

- Ans ☒ 1. Husband's father's sister
- ☒ 2. Husband's mother's sister
- ☒ 3. Husband's father's mother
- ☒ 4. Husband's mother's mother

Question ID : 6306801026844
Option 1 ID : 6306804030046
Option 2 ID : 6306804030048
Option 3 ID : 6306804030047
Option 4 ID : 6306804030049
Status : Answered
Chosen Option : 1

Q.26 In the number 6975184, what is the least common multiple (LCM) of the digits at the 1st, 3rd, 5th and 7th positions when counted from the left end?

- Ans ☒ 1. 84
- ☒ 2. 252
- ☒ 3. 378
- ☒ 4. 42

Question ID : 630680264612
Option 1 ID : 6306801027953
Option 2 ID : 6306801027952
Option 3 ID : 6306801027954
Option 4 ID : 6306801027951
Status : Answered
Chosen Option : 1

Q.27 Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s).

Statements:

Some shapes are circle.
Some shapes are long.
No triangle is long.

Conclusions:

(I) Some circles are long.
(II) All shapes are triangle.

Ans ☒ 1. Only conclusion (I) follows.

☒ 2. Both conclusions (I) and (II) follows.

☒ 3. Only conclusion (II) follows.

☒ 4. Neither conclusion (I) nor (II) follows.

Question ID : 630680499276

Option 1 ID : 6306801950768

Option 2 ID : 6306801950771

Option 3 ID : 6306801950769

Option 4 ID : 6306801950770

Status : Answered

Chosen Option : 1

Q.28 Read the given statements and conclusions carefully. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which conclusion/s logically follow/s from the given statements.

Statements:

All coats are belts.
All belts are watches.
All watches are jackets.

Conclusions:

(I) All coats are watches.
(II) All belts are jackets.

Ans ☒ 1. Both conclusions I and II follow.

☒ 2. Only conclusion II follows.

☒ 3. Neither conclusion I nor II follows.

☒ 4. Only conclusion I follows.

Question ID : 630680421200

Option 1 ID : 6306801643277

Option 2 ID : 6306801643276

Option 3 ID : 6306801643278

Option 4 ID : 6306801643275

Status : Answered

Chosen Option : 1

Q.29 DFGJ is related to EHJN in a certain way based on the English alphabetical order. In the same way, GLPV is related to HNSZ. To which of the following is KTBL related, following the same logic?

Ans ☒ 1. LKIU

☒ 2. LVWP

☒ 3. LVEP

☒ 4. LPTY

Question ID : 6306801026847

Option 1 ID : 6306804030053

Option 2 ID : 6306804030050

Option 3 ID : 6306804030051

Option 4 ID : 6306804030052

Status : Answered

Chosen Option : 3

Question ID : 630680908032
Option 1 ID : 6306803557201
Option 2 ID : 6306803557200
Option 3 ID : 6306803557198
Option 4 ID : 6306803557199
Status : Answered
Chosen Option : 4

Ans ☒ 1.3
☐ 2.5
☐ 3.2
☐ 4.4

Question ID : 630680908103
Option 1 ID : 6306803557479
Option 2 ID : 6306803557481
Option 3 ID : 6306803557478
Option 4 ID : 6306803557480
Status : Answered
Chosen Option : 2

Ans ☒ 1. 4
☒ 2. 8
☒ 3. 9
☒ 4. 6

Question ID : 6306801026883
Option 1 ID : 6306804030173
Option 2 ID : 6306804030171
Option 3 ID : 6306804030172
Option 4 ID : 6306804030170
Status : Answered
Chosen Option : 2

Ans  1. 0
 2. 2
 3. 3
 4. 1

Question ID : 630680908120
Option 1 ID : 6306803557554
Option 2 ID : 6306803557556
Option 3 ID : 6306803557557
Option 4 ID : 6306803557555
Status : Answered
Chosen Option : 4

Q.34 Three of the following four letter-cluster-pairs are alike in a certain way and thus form a group. Which letter-cluster-pair does NOT belong to that group?

- Ans
- ☒ 1. AD-MP
 - ☒ 2. KN-WZ
 - ☒ 3. GJ-RV
 - ☒ 4. EH-QT

Question ID : 630680187259
Option 1 ID : 630680725986
Option 2 ID : 630680725987
Option 3 ID : 630680725988
Option 4 ID : 630680725985
Status : Answered
Chosen Option : 3

Section : Domain Knowledge

Q.1 Which of the following statements about the single parameter model is accurate?

- Ans
- ☒ 1. It requires extensive experimental data to validate its predictions.
 - ☒ 2. It provides a detailed mechanistic understanding of the reaction pathways.
 - ☒ 3. It incorporates the effects of temperature and pressure explicitly in its formulation.
 - ☒ 4. It is particularly useful for quick approximations of reactor performance.

Question ID : 630680983979
Option 1 ID : 6306803859980
Option 2 ID : 6306803859978
Option 3 ID : 6306803859977
Option 4 ID : 6306803859979
Status : Answered
Chosen Option : 2

Q.2 Convection heat transfer involves:

- Ans
- ☒ 1. heat transfer due to fluid motion
 - ☒ 2. heat transfer by electromagnetic radiation
 - ☒ 3. heat transfer within a solid
 - ☒ 4. heat transfer in a vacuum

Question ID : 630680974139
Option 1 ID : 6306803820217
Option 2 ID : 6306803820216
Option 3 ID : 6306803820215
Option 4 ID : 6306803820218
Status : Answered
Chosen Option : 1

Q.3 In a process with 5 unknowns and 3 independent equations, what is the degree of freedom?

- Ans
- ☒ 1. 2
 - ☒ 2. 8
 - ☒ 3. 5
 - ☒ 4. 3

Question ID : 630680970665
Option 1 ID : 6306803806919
Option 2 ID : 6306803806922
Option 3 ID : 6306803806921
Option 4 ID : 6306803806920
Status : Answered
Chosen Option : 1

Q.4 Who is responsible for providing Safety Data Sheets (SDS) to customers?

- Ans
- ☒ 1. Health and safety inspectors
 - ☒ 2. Consumers
 - ☒ 3. Chemical suppliers, including manufacturers, importers, distributors, wholesalers and retailers
 - ☒ 4. Environmental agencies

Question ID : 630680959866
Option 1 ID : 6306803763396
Option 2 ID : 6306803763394
Option 3 ID : 6306803763393
Option 4 ID : 6306803763395
Status : Answered
Chosen Option : 3

Q.5 When depreciation is calculated using the straight-line method, how does the depreciation change over the years?

- Ans
- ☒ 1. Increases with every year
 - ☒ 2. Is simplified and consistent
 - ☒ 3. Remains constant
 - ☒ 4. Decreases with every year

Question ID : 630680959297
Option 1 ID : 6306803760977
Option 2 ID : 6306803760979
Option 3 ID : 6306803760976
Option 4 ID : 6306803760978
Status : Answered
Chosen Option : 3

Q.6 In an ideal Plug Flow Reactor (PFR), the concentration of reactants:

- Ans
- ☒ 1. is the same as in a Continuous Stirred Tank Reactor (CSTR)
 - ☒ 2. increases along the length of the reactor
 - ☒ 3. decreases along the length of the reactor
 - ☒ 4. remains constant throughout the reactor

Question ID : 630680983142
Option 1 ID : 6306803856530
Option 2 ID : 6306803856528
Option 3 ID : 6306803856529
Option 4 ID : 6306803856527
Status : Answered
Chosen Option : 3

Q.7 Which of the following statements regarding shell and tube heat exchangers is/are correct?

Statement A: The primary purpose of baffles in a shell and tube heat exchanger is to increase the tube-side fluid velocity.
Statement B: The Log Mean Temperature Difference (LMTD) is a key parameter used to calculate the heat transfer rate in a heat exchanger.

- Ans
- ☒ 1. Both Statement A and Statement B
 - ☒ 2. Only Statement A
 - ☒ 3. Only Statement B
 - ☒ 4. Neither Statement A nor Statement B

Question ID : 630680974503
Option 1 ID : 6306803821710
Option 2 ID : 6306803821708
Option 3 ID : 6306803821709
Option 4 ID : 6306803821711
Status : Answered
Chosen Option : 1

Q.8 Which of the following statements is true regarding the relationship between fires and explosions?

- Ans
- ☒ 1. Fires never result from explosions.
 - ☒ 2. Explosions can result from fires.
 - ☒ 3. Explosions always precede fires.
 - ☒ 4. Fires release energy more rapidly than explosions.

Question ID : 630680959862
Option 1 ID : 6306803763377
Option 2 ID : 6306803763378
Option 3 ID : 6306803763380
Option 4 ID : 6306803763379
Status : Marked For Review
Chosen Option : 2

Q.9 Polyesters are typically synthesised through which polymerisation process?

- Ans
- ☒ 1. Addition polymerisation
 - ☒ 2. Condensation polymerisation
 - ☒ 3. Emulsion polymerisation
 - ☒ 4. Suspension polymerisation

Question ID : 630680959102
Option 1 ID : 6306803760205
Option 2 ID : 6306803760204
Option 3 ID : 6306803760206
Option 4 ID : 6306803760207
Status : Answered
Chosen Option : 1

Q.10 Which type of fire extinguisher is commonly used for Class B fires involving flammable liquids?

- Ans ☒ 1. Water
- ☒ 2. Carbon dioxide
- ☒ 3. Dry powder
- ☒ 4. Foam

Question ID : 630680959861
 Option 1 ID : 6306803763373
 Option 2 ID : 6306803763376
 Option 3 ID : 6306803763375
 Option 4 ID : 6306803763374
 Status : Answered
 Chosen Option : 2

Q.11 Cross-validation is a very important step performed while building machine learning models for quality control in chemical engineering. Which of the following statements is true about the cross-validation process?

- Ans ☒ 1. Cross-validation is performed to reduce the size of the input data used by the model.
- ☒ 2. Cross-validation is performed to ensure that the input data (raw material) used in the chemical process is valid and is of accurate quantity.
- ☒ 3. Cross-validation is performed to ensure that the model performs well on the unseen data.
- ☒ 4. The speed of the model increases after performing cross-validation.

Question ID : 630680978863
 Option 1 ID : 6306803839440
 Option 2 ID : 6306803839438
 Option 3 ID : 6306803839439
 Option 4 ID : 6306803839437
 Status : Answered
 Chosen Option : 2

Q.12 Which energy carrier is mentioned as a low-emissions alternative in decarbonisation efforts?

- Ans ☒ 1. Diesel
- ☒ 2. Coal
- ☒ 3. Natural gas
- ☒ 4. Hydrogen

Question ID : 630680959867
 Option 1 ID : 6306803763400
 Option 2 ID : 6306803763398
 Option 3 ID : 6306803763399
 Option 4 ID : 6306803763397
 Status : Answered
 Chosen Option : 4

Q.13 Navier-Stokes equation can be written as:

- Ans ☒ 1. $\frac{\rho Dv}{Dt} = + \nabla p - \mu \nabla^2 v + \rho g$
- ☒ 2. $\frac{\rho Dv}{Dt} = - \nabla p + \mu \nabla^2 v + \rho g$
- ☒ 3. $\frac{\rho Dv}{Dt} = - \nabla p - \mu \nabla^2 v + \rho g$
- ☒ 4. $\frac{\rho Dv}{Dt} = - \nabla p + \mu \nabla^2 v - \rho g$

Question ID : 630680959922
 Option 1 ID : 6306803763602
 Option 2 ID : 6306803763601
 Option 3 ID : 6306803763604
 Option 4 ID : 6306803763603
 Status : Answered
 Chosen Option : 2

Q.14 What is a common characteristic of renewable energy sources?

- Ans
- ☒ 1. They are finite and will eventually be depleted.
 - ☒ 2. They require human interaction to release energy.
 - ☒ 3. They involve energy that is already passing through the environment as a current or flow.
 - ☒ 4. They are primarily derived from chemical reactions.

Question ID : 630680959868
Option 1 ID : 6306803763403
Option 2 ID : 6306803763404
Option 3 ID : 6306803763401
Option 4 ID : 6306803763402
Status : Answered
Chosen Option : 3

Q.15 Activity coefficients are used to account for:

- Ans
- ☒ 1. non-ideal solution behaviour
 - ☒ 2. phase equilibrium of pure substances
 - ☒ 3. heat capacity of gases
 - ☒ 4. ideal gas behaviour

Question ID : 630680971420
Option 1 ID : 6306803809696
Option 2 ID : 6306803809697
Option 3 ID : 6306803809698
Option 4 ID : 6306803809695
Status : Answered
Chosen Option : 1

Q.16 Which of the following characteristics is indicative of an underdamped second-order system's response to a step input?

- Ans
- ☒ 1. Immediate and smooth approach to the new steady state
 - ☒ 2. Oscillations with a decaying amplitude over time
 - ☒ 3. A response that reaches steady state with no oscillations
 - ☒ 4. No overshoot and a slow return to steady state

Question ID : 630680952560
Option 1 ID : 6306803734876
Option 2 ID : 6306803734875
Option 3 ID : 6306803734874
Option 4 ID : 6306803734873
Status : Answered
Chosen Option : 2

Q.17 Which of the following statements regarding stage efficiencies in mass transfer operations is/are correct?

Statement A: The Murphree stage efficiency is a measure of the actual performance of a stage compared to the theoretical performance.
Statement B: The overall stage efficiency is always higher than the Murphree stage efficiency.

- Ans
- ☒ 1. Neither Statement A nor Statement B
 - ☒ 2. Only Statement A
 - ☒ 3. Only Statement B
 - ☒ 4. Both Statement A and Statement B

Question ID : 630680974752
Option 1 ID : 6306803822817
Option 2 ID : 6306803822816
Option 3 ID : 6306803822814
Option 4 ID : 6306803822815
Status : Answered
Chosen Option : 2

Q.18 Which of the following statements about the kinetics of heterogeneous catalytic reactions is true?

- Ans
- ☒ 1. The reaction rate depends on the adsorption of reactants on the catalyst surface.
 - ☒ 2. The rate of reaction is independent of the catalyst surface area.
 - ☒ 3. These reactions occur in a single phase.
 - ☒ 4. Heterogeneous catalytic reactions do not involve mass transfer limitations.

Question ID : 630680983957
Option 1 ID : 6306803859875
Option 2 ID : 6306803859874
Option 3 ID : 6306803859873
Option 4 ID : 6306803859876
Status : Answered
Chosen Option : 1

Q.19 Which of the following expressions is correct?

- Ans ☒ 1. $\text{Rate} = (\text{Driving Force}) / (\text{Resistance})$
☒ 2. $\text{Rate} = (\text{Driving Force}) + (\text{Resistance})$
☒ 3. $\text{Rate} = (\text{Resistance}) / (\text{Driving Force})$
☒ 4. $\text{Rate} = (\text{Driving Force}) \times (\text{Resistance})$

Question ID : 630680959917
 Option 1 ID : 6306803763581
 Option 2 ID : 6306803763584
 Option 3 ID : 6306803763582
 Option 4 ID : 6306803763583
 Status : Answered
 Chosen Option : 1

Q.20 What effect does increasing the Derivative gain (K_d) have on a PID controller?

- Ans ☒ 1. It increases the steady-state error.
☒ 2. It eliminates the offset completely.
☒ 3. It decreases the response time of the system.
☒ 4. It reduces the overshoot in the system response.

Question ID : 630680952581
 Option 1 ID : 6306803734955
 Option 2 ID : 6306803734956
 Option 3 ID : 6306803734954
 Option 4 ID : 6306803734953
 Status : Answered
 Chosen Option : 3

Q.21 According to the collision theory, the rate of a chemical reaction depends on:

- Ans ☒ 1. the temperature and the molecular structure of the reactants
☒ 2. the temperature and the activation energy of the reactants
☒ 3. the concentration and the molecular weight of the reactants
☒ 4. the activation energy and the molecular weight of the reactants

Question ID : 630680982999
 Option 1 ID : 6306803855966
 Option 2 ID : 6306803855963
 Option 3 ID : 6306803855964
 Option 4 ID : 6306803855965
 Status : Answered
 Chosen Option : 2

Q.22 Which of the following statements regarding Fick's law is/are correct?

Statement A: Fick's first law states that the mass flux due to diffusion is proportional to the concentration gradient.
 Statement B: Fick's second law describes the change in concentration with respect to time and space.

- Ans ☒ 1. Only Statement A
☒ 2. Both Statement A and Statement B
☒ 3. Neither Statement A nor Statement B
☒ 4. Only Statement B

Question ID : 630680974657
 Option 1 ID : 6306803822380
 Option 2 ID : 6306803822382
 Option 3 ID : 6306803822383
 Option 4 ID : 6306803822381
 Status : Answered
 Chosen Option : 2

Q.23 The fertiliser composition expressed as 10-5-20 represents the percentage by weight of which nutrients?

- Ans ☒ 1. 10 wt.% N, 5 wt.% P_2O_5 , 20 wt.% K_2O
☒ 2. 10 wt.% N_2O , 5 wt.% P_2O_5 , 20 wt.% K_2O
☒ 3. 10 wt.% N, 5 wt.% P_2O_5 , 20 wt.% K_2O
☒ 4. 10 wt.% N_2O , 5 wt.% P_2O_5 , 20 wt.% K_2O

Question ID : 630680959154
 Option 1 ID : 6306803760411
 Option 2 ID : 6306803760409
 Option 3 ID : 6306803760408
 Option 4 ID : 6306803760410
 Status : Not Answered
 Chosen Option : --

Q.24 In a closed system undergoing a thermodynamic cycle, the net work done by the system is equal to:

- Ans
- ☐ 1. the change in internal energy of the system
 - ☐ 2. the change in entropy of the system
 - ☐ 3. the change in enthalpy of the system
 - ☒ 4. the net heat added to the system

Question ID : 630680971049
Option 1 ID : 6306803808436
Option 2 ID : 6306803808438
Option 3 ID : 6306803808437
Option 4 ID : 6306803808435
Status : Answered
Chosen Option : 4

Q.25 In a circular economy, what replaces the 'end-of-life' concept for materials?

- Ans
- ☐ 1. Dispose and forget
 - ☐ 2. Store indefinitely
 - ☒ 3. Reuse, recycle, recover
 - ☐ 4. Export to other countries

Question ID : 630680959879
Option 1 ID : 6306803763442
Option 2 ID : 6306803763443
Option 3 ID : 6306803763441
Option 4 ID : 6306803763444
Status : Answered
Chosen Option : 3

Q.26 Which of the following best describes the primary purpose of Aspen Plus® and Aspen Hysys®?

- Ans
- ☐ 1. Analysing market trends for process optimisation
 - ☒ 2. Predicting process behaviour and unit operations
 - ☐ 3. Generating experimental data for chemical transformations
 - ☐ 4. Performing sensitivity analysis on equipment sizes

Question ID : 630680963493
Option 1 ID : 6306803778530
Option 2 ID : 6306803778527
Option 3 ID : 6306803778529
Option 4 ID : 6306803778528
Status : Marked For Review
Chosen Option : 1

Q.27 In steady state heat transfer through a cylindrical pipe, the temperature distribution is:

- Ans
- ☐ 1. linear in the radial direction
 - ☒ 2. logarithmic in the radial direction
 - ☐ 3. exponential in the radial direction
 - ☐ 4. parabolic in the radial direction

Question ID : 630680974199
Option 1 ID : 6306803820449
Option 2 ID : 6306803820451
Option 3 ID : 6306803820450
Option 4 ID : 6306803820452
Status : Answered
Chosen Option : 2

Q.28 In a heat exchanger, the log mean temperature difference (LMTD) is used to:

- Ans
- ☐ 1. find the Reynolds number for fluid flow
 - ☐ 2. determine the overall heat transfer coefficient
 - ☐ 3. calculate the thermal conductivity of materials
 - ☒ 4. calculate the heat transfer area required

Question ID : 630680974306
Option 1 ID : 6306803820920
Option 2 ID : 6306803820918
Option 3 ID : 6306803820917
Option 4 ID : 6306803820919
Status : Answered
Chosen Option : 4

Q.29 What hazardous property of chemicals can cause an immediate reaction when exposed to a flame or spark?

- Ans
- ☒ 1. Toxicity
 - ☒ 2. Corrosivity
 - ☒ 3. Flammability
 - ☒ 4. Reactivity

Question ID : 630680959863
Option 1 ID : 6306803763382
Option 2 ID : 6306803763383
Option 3 ID : 6306803763381
Option 4 ID : 6306803763384
Status : Answered
Chosen Option : 3

Q.30 What is the primary function of an Integral (I) action in a PI controller?

- Ans
- ☒ 1. To differentiate the error signal with respect to time
 - ☒ 2. To eliminate steady-state error completely
 - ☒ 3. To provide a control action that is proportional to the error signal
 - ☒ 4. To integrate the error signal over time

Question ID : 630680952570
Option 1 ID : 6306803734916
Option 2 ID : 6306803734913
Option 3 ID : 6306803734914
Option 4 ID : 6306803734915
Status : Answered
Chosen Option : 2

Q.31 Which of the following statements regarding HTU (Height of Transfer Unit) and NTU (Number of Transfer Units) concepts in mass transfer is/are correct?
Statement A: The Height of a Transfer Unit (HTU) is a measure of the height required for one transfer unit of mass to be transferred between phases.
Statement B: The Number of Transfer Units (NTU) represents the efficiency of the mass transfer process and is defined as the ratio of the actual mass transfer to the maximum possible mass transfer.

- Ans
- ☒ 1. Both Statement A and Statement B are incorrect.
 - ☒ 2. Both Statement A and Statement B are correct.
 - ☒ 3. Only Statement B is correct.
 - ☒ 4. Only Statement A is correct.

Question ID : 630680974802
Option 1 ID : 6306803823080
Option 2 ID : 6306803823079
Option 3 ID : 6306803823078
Option 4 ID : 6306803823077
Status : Answered
Chosen Option : 2

Q.32 In the manufacturing of plastic bottles, which specific type of polyethylene is typically used?

- Ans
- ☒ 1. High-Density Polyethylene (HDPE)
 - ☒ 2. Linear Low-Density Polyethylene (LLDPE)
 - ☒ 3. Linear High-Density Polyethylene (LHDPE)
 - ☒ 4. Low-Density Polyethylene (LDPE)

Question ID : 630680959080
Option 1 ID : 6306803760116
Option 2 ID : 6306803760118
Option 3 ID : 6306803760119
Option 4 ID : 6306803760117
Status : Answered
Chosen Option : 2

Q.33 The payout period is defined as the minimum length of time necessary to:

- Ans
- ☒ 1. double the original capital investment
 - ☒ 2. recover the original capital investment in the form of cash flow
 - ☒ 3. achieve the maximum profit
 - ☒ 4. reach break-even without considering any costs

Question ID : 630680959358
Option 1 ID : 6306803761217
Option 2 ID : 6306803761216
Option 3 ID : 6306803761218
Option 4 ID : 6306803761219
Status : Answered
Chosen Option : 2

Q.34 Select the option that is correct regarding the following two statements labelled Assertion (A) and Reason (R).
Assertion (A): The rate of a chemical reaction increases exponentially with an increase in temperature.
Reason (R): According to the Arrhenius equation, the rate constant k is directly proportional to the exponential of the negative activation energy divided by the product of the gas constant and temperature.

- Ans
- ☐ 1. A is true, but R is false.
 - ☐ 2. A is false, but R is true.
 - ☒ 3. Both A and R are true, and R is the correct explanation of A.
 - ☐ 4. Both A and R are true, but R is not the correct explanation of A.

Question ID : 630680983087
Option 1 ID : 6306803856308
Option 2 ID : 6306803856309
Option 3 ID : 6306803856306
Option 4 ID : 6306803856307
Status : Marked For Review
Chosen Option : 2

Q.35 In a non-ideal reactor, which concept is often used to analyse the residence time distribution (RTD) of the fluid?

- Ans
- ☒ 1. The E-curve and F-curve
 - ☐ 2. The Langmuir-Hinshelwood mechanism
 - ☐ 3. The Gibbs free energy change
 - ☐ 4. The Arrhenius equation

Question ID : 630680983162
Option 1 ID : 6306803856608
Option 2 ID : 6306803856609
Option 3 ID : 6306803856610
Option 4 ID : 6306803856607
Status : Answered
Chosen Option : 1

Q.36 What is the primary benefit of using a first-principles model in process control?

- Ans
- ☒ 1. It provides a detailed, mechanistic understanding of the process.
 - ☐ 2. It always results in a simpler model than empirical modelling.
 - ☐ 3. It requires no prior knowledge of the process dynamics.
 - ☐ 4. It can be easily linearised for any operating point.

Question ID : 630680952556
Option 1 ID : 6306803734858
Option 2 ID : 6306803734860
Option 3 ID : 6306803734857
Option 4 ID : 6306803734859
Status : Answered
Chosen Option : 2

Q.37 In a fluid at rest, the pressure at any given point is:

- Ans
- ☐ 1. independent of density
 - ☐ 2. dependent on flow rate
 - ☐ 3. dependent on viscosity
 - ☒ 4. independent of direction

Question ID : 630680972356
Option 1 ID : 6306803813206
Option 2 ID : 6306803813205
Option 3 ID : 6306803813204
Option 4 ID : 6306803813203
Status : Answered
Chosen Option : 4

Q.38 What is the primary advantage of using a 4-20 mA current loop for transmitting process variable signals?

- Ans
- ☐ 1. It is less expensive compared to voltage signals.
 - ☐ 2. It is immune to electromagnetic interference.
 - ☐ 3. It allows for bidirectional communication.
 - ☒ 4. It can transmit signals over long distances without signal degradation.

Question ID : 630680952545
Option 1 ID : 6306803734814
Option 2 ID : 6306803734815
Option 3 ID : 6306803734816
Option 4 ID : 6306803734813
Status : Answered
Chosen Option : 4

Q.39 What is/are the primary feedstocks for the petrochemical industry?

- Ans
- ☒ 1. Petroleum products
 - ☐ 2. Coal
 - ☐ 3. Fertilisers
 - ☐ 4. Biomass

Question ID : 630680959039
Option 1 ID : 6306803759942
Option 2 ID : 6306803759944
Option 3 ID : 6306803759945
Option 4 ID : 6306803759943
Status : Answered
Chosen Option : 1

Q.40 The Darcy-Weisbach equation relates the head loss due to friction in a pipe to:

- Ans
- ☒ 1. the friction factor, pipe length and the velocity head
 - ☐ 2. the Reynolds number and the viscosity of the fluid
 - ☐ 3. the velocity of the fluid and the pipe diameter
 - ☐ 4. the flow rate and pipe length

Question ID : 630680973157
Option 1 ID : 6306803816401
Option 2 ID : 6306803816402
Option 3 ID : 6306803816400
Option 4 ID : 6306803816399
Status : Answered
Chosen Option : 1

Q.41 What are the ideal characteristics of an alternative energy technology?

- Ans
- ☐ 1. Limited, non-renewable, versatile, clean, economical
 - ☐ 2. Abundant, expensive, non-renewable, versatile, clean
 - ☒ 3. Clean, unlimited, renewable, versatile, economical
 - ☐ 4. Limited, renewable, expensive, clean, versatile

Question ID : 630680959870
Option 1 ID : 6306803763412
Option 2 ID : 6306803763411
Option 3 ID : 6306803763409
Option 4 ID : 6306803763410
Status : Answered
Chosen Option : 3

Q.42 Which of the following statements regarding the drying process in mass transfer operations is/are correct?

Statement A: In the constant rate period of drying, the rate of moisture removal is limited by the rate of heat transfer to the material being dried.
Statement B: During the falling rate period of drying, the rate of moisture removal is primarily controlled by the diffusion of moisture from within the material to its surface.

- Ans
- ☐ 1. Both Statement A and Statement B are incorrect.
 - ☐ 2. Only Statement B is correct.
 - ☒ 3. Both Statement A and Statement B are correct.
 - ☐ 4. Only Statement A is correct.

Question ID : 6306801025142
Option 1 ID : 6306804023278
Option 2 ID : 6306804023276
Option 3 ID : 6306804023277
Option 4 ID : 6306804023275
Status : Not Answered
Chosen Option : --

Q.43 Which of the following is a primary use of tie components in chemical process calculations?

- Ans
- ☐ 1. To calculate the heat transfer coefficient
 - ☐ 2. To design the dimensions of a distillation column
 - ☒ 3. To simplify the material balance equations for a system
 - ☐ 4. To determine the energy balance for a reactor

Question ID : 630680970541
Option 1 ID : 6306803806427
Option 2 ID : 6306803806430
Option 3 ID : 6306803806428
Option 4 ID : 6306803806429
Status : Answered
Chosen Option : 3

Q.44 The sphericity of a particle is defined as:

- Ans
- ☐ 1. the ratio of the particle's volume to its surface area
 - ☐ 2. the ratio of the volume of a sphere of the same surface area to the actual volume of the particle
 - ☐ 3. the ratio of the particle's surface area to its volume
 - ☒ 4. the ratio of the surface area of a sphere of the same volume to the actual surface area of the particle

Question ID : 630680973451
Option 1 ID : 6306803817570
Option 2 ID : 6306803817568
Option 3 ID : 6306803817569
Option 4 ID : 6306803817567
Status : Answered
Chosen Option : 4

Q.45 When predicting the vapour-liquid equilibrium (VLE) of a system, which of the following models might be used?

- Ans
- ☐ 1. Bernoulli's Equation
 - ☐ 2. Nernst Equation
 - ☐ 3. Fourier's Law
 - ☒ 4. Raoult's Law

Question ID : 630680972232
Option 1 ID : 6306803812708
Option 2 ID : 6306803812709
Option 3 ID : 6306803812710
Option 4 ID : 6306803812707
Status : Answered
Chosen Option : 4

Q.46 Consider a hyperplane in 'n' dimension. The equation of the hyperplane is given as follows:

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + w_0 = 0$$

What do the values ($w_1, w_2, w_3, \dots, w_n$) represent in the equation?

- Ans
- ☐ 1. They represent vectors that are parallel to the hyperplane.
 - ☒ 2. They represent components of the vector that is normal to the hyperplane.
 - ☐ 3. They represent tangent vectors to the hyperplane.
 - ☐ 4. They represent components of 'n' dimensional vector that passes through the origin.

Question ID : 630680969265
Option 1 ID : 6306803801563
Option 2 ID : 6306803801564
Option 3 ID : 6306803801566
Option 4 ID : 6306803801565
Status : Answered
Chosen Option : 2

Q.47 Which class of fire involves solid carbonaceous materials such as fibres and plastics?

- Ans
- ☐ 1. Class B
 - ☐ 2. Class C
 - ☐ 3. Class D
 - ☒ 4. Class A

Question ID : 630680959860
Option 1 ID : 6306803763370
Option 2 ID : 6306803763371
Option 3 ID : 6306803763372
Option 4 ID : 6306803763369
Status : Not Answered
Chosen Option : --

Q.48 For a first-order reaction, the rate of reaction is proportional to:

- Ans ☒ 1. the concentration of the product
- ☒ 2. the concentration of the reactant
- ☒ 3. the inverse of the concentration of the reactant
- ☒ 4. the square of the concentration of the reactant

Question ID : 630680983117
 Option 1 ID : 6306803856430
 Option 2 ID : 6306803856428
 Option 3 ID : 6306803856429
 Option 4 ID : 6306803856427
 Status : Answered
 Chosen Option : 2

Q.49 Which statement best describes the use of iterative calculation in optimisation simulators?

- Ans ☒ 1. Iterative calculations are primarily used in external loops to achieve convergence.
- ☒ 2. Iterative calculations are used to adjust design specifications and achieve convergence.
- ☒ 3. Iterative calculations are not essential in optimisation simulators.
- ☒ 4. Iterative calculations are only applied in recycle loops to specify design criteria.

Question ID : 630680959891
 Option 1 ID : 6306803763488
 Option 2 ID : 6306803763485
 Option 3 ID : 6306803763486
 Option 4 ID : 6306803763487
 Status : Answered
 Chosen Option : 2

Q.50 Which of the following documents provides information on the hazards of a chemical substance or mixture and includes advice on precautions during handling?

- Ans ☒ 1. Technical Data Sheet (TDS)
- ☒ 2. Material Safety Data Sheet (MSDS)
- ☒ 3. Safety Compliance Sheet (SCS)
- ☒ 4. Hazard Assessment Report (HAR)

Question ID : 630680959865
 Option 1 ID : 6306803763390
 Option 2 ID : 6306803763389
 Option 3 ID : 6306803763391
 Option 4 ID : 6306803763392
 Status : Answered
 Chosen Option : 2

Q.51 Which of the following best describes a Newtonian fluid?

- Ans ☒ 1. A fluid that behaves as an elastic solid under stress
- ☒ 2. A fluid whose viscosity stays constant with shear rate
- ☒ 3. A fluid whose viscosity decreases with shear rate
- ☒ 4. A fluid whose viscosity increases with shear rate

Question ID : 630680972429
 Option 1 ID : 6306803813498
 Option 2 ID : 6306803813497
 Option 3 ID : 6306803813495
 Option 4 ID : 6306803813496
 Status : Answered
 Chosen Option : 2

Q.52 Which method of carbon storage is most likely to prevent CO₂ from escaping indefinitely?

- Ans ☒ 1. Solid storage using calcium oxide
- ☒ 2. Underground storage in depleted oil wells
- ☒ 3. Underground storage in depleted natural gas wells
- ☒ 4. Ocean storage as gaseous CO₂ near the surface

Question ID : 630680959875
 Option 1 ID : 6306803763427
 Option 2 ID : 6306803763425
 Option 3 ID : 6306803763428
 Option 4 ID : 6306803763426
 Status : Not Answered
 Chosen Option : --

Q.53 During the humidification process, the addition of moisture to air is typically achieved by:

- Ans
- ☒ 1. cooling the air below its dew point
 - ☒ 2. heating the air without adding moisture
 - ☒ 3. passing the air through a water spray
 - ☒ 4. using a desiccant material

Question ID : 6306801025141
Option 1 ID : 6306804023274
Option 2 ID : 6306804023271
Option 3 ID : 6306804023272
Option 4 ID : 6306804023273
Status : Answered
Chosen Option : 1

Q.54 Which of the following options describes the dynamic behaviour of a first-order transducer when subjected to a step input?

- Ans
- ☒ 1. Linear increase to the final value
 - ☒ 2. Instantaneous response with no delay
 - ☒ 3. Exponential approach to the final value with a characteristic time constant
 - ☒ 4. Oscillatory response with damping

Question ID : 630680952551
Option 1 ID : 6306803734840
Option 2 ID : 6306803734837
Option 3 ID : 6306803734838
Option 4 ID : 6306803734839
Status : Answered
Chosen Option : 3

Q.55 In dimensional analysis, the Buckingham π theorem is used to:

- Ans
- ☒ 1. solve linear differential equations
 - ☒ 2. convert different units of measurement
 - ☒ 3. predict the viscosity of a fluid
 - ☒ 4. find dimensionless groups in problems

Question ID : 630680973273
Option 1 ID : 6306803816861
Option 2 ID : 6306803816859
Option 3 ID : 6306803816862
Option 4 ID : 6306803816860
Status : Answered
Chosen Option : 4

Q.56 Which type of evaporator is best suited for materials sensitive to high temperatures?

- Ans
- ☒ 1. Climbing film evaporator
 - ☒ 2. Natural circulation evaporator
 - ☒ 3. Forced circulation evaporator
 - ☒ 4. Falling film evaporator

Question ID : 630680974271
Option 1 ID : 6306803820764
Option 2 ID : 6306803820761
Option 3 ID : 6306803820762
Option 4 ID : 6306803820763
Status : Answered
Chosen Option : 4

Q.57 What is the primary purpose of environmental monitoring?

- Ans
- ☒ 1. To characterise and monitor the quality of the environment
 - ☒ 2. To increase the economic value of natural resources
 - ☒ 3. To reduce the cost of environmental projects
 - ☒ 4. To develop new industrial projects

Question ID : 630680959869
Option 1 ID : 6306803763405
Option 2 ID : 6306803763406
Option 3 ID : 6306803763408
Option 4 ID : 6306803763407
Status : Answered
Chosen Option : 1

Q.58 Boundary layer separation occurs when:

- Ans ☒ 1. fluid velocity exceeds speed of sound
☒ 2. the Reynolds number is very low
☒ 3. adverse pressure gradient reverses flow
☒ 4. the fluid is highly viscous

Question ID : 630680973378
Option 1 ID : 6306803817275
Option 2 ID : 6306803817277
Option 3 ID : 6306803817276
Option 4 ID : 6306803817278
Status : Answered
Chosen Option : 3

Q.59 In the design of double pipe heat exchangers, which configuration provides the maximum heat transfer efficiency?

- Ans ☒ 1. Parallel flow
☒ 2. Counterflow
☒ 3. Radial flow
☒ 4. Crossflow

Question ID : 630680974462
Option 1 ID : 6306803821544
Option 2 ID : 6306803821545
Option 3 ID : 6306803821547
Option 4 ID : 6306803821546
Status : Answered
Chosen Option : 2

Q.60 Which of the following materials is commonly used as an adsorbent in dehumidification processes?

- Ans ☒ 1. Activated carbon
☒ 2. Sodium bicarbonate
☒ 3. Calcium chloride
☒ 4. Silica gel

Question ID : 630680977022
Option 1 ID : 6306803832164
Option 2 ID : 6306803832166
Option 3 ID : 6306803832165
Option 4 ID : 6306803832163
Status : Answered
Chosen Option : 4

Q.61 'Linear Regression' is a machine learning technique that is used for making predictions. What is the use of 'Linear Regression' in the context of 'Emission Control' of the chemical engineering process?

- Ans ☒ 1. To create clusters of emission sources
☒ 2. To perform classification of pollutants
☒ 3. To predict the quantity of emission
☒ 4. To reduce dimensionality of the data set

Question ID : 630680978949
Option 1 ID : 6306803839778
Option 2 ID : 6306803839777
Option 3 ID : 6306803839780
Option 4 ID : 6306803839779
Status : Answered
Chosen Option : 3

Q.62 What does 'K' represent in K-Nearest Neighbor classification algorithm?

- Ans ☒ 1. Number of nearest neighbours to consider
☒ 2. Number of columns (features) in the data set
☒ 3. Number of outliers in the data set
☒ 4. Number of rows (data points) in the data set

Question ID : 630680978871
Option 1 ID : 6306803839471
Option 2 ID : 6306803839469
Option 3 ID : 6306803839472
Option 4 ID : 6306803839470
Status : Marked For Review
Chosen Option : 3

Q.63 Linear Low-Density Polyethylene (LLDPE) is produced under:

- Ans ☒ 1. low pressure
- ☒ 2. high pressure
- ☒ 3. moderate pressure
- ☒ 4. very low pressure

Question ID : 630680959065
Option 1 ID : 6306803760056
Option 2 ID : 6306803760059
Option 3 ID : 6306803760057
Option 4 ID : 6306803760058
Status : Answered
Chosen Option : 2

Q.64 The pressure drop in a pipeline system is influenced by:

- Ans ☒ 1. the gravitational acceleration only
- ☒ 2. the fluid's temperature and density
- ☒ 3. the ambient atmospheric pressure
- ☒ 4. the pipe's roughness, length, diameter and flow rate

Question ID : 630680973311
Option 1 ID : 6306803817014
Option 2 ID : 6306803817011
Option 3 ID : 6306803817013
Option 4 ID : 6306803817012
Status : Answered
Chosen Option : 4

Q.65 Which of the following statements is true about semi-supervised learning in machine learning?

- Ans ☒ 1. The data used in semi-supervised learning is unlabelled and hence does not have the output label column.
- ☒ 2. The data used in semi-supervised learning is a mix of labelled and unlabelled data.
- ☒ 3. K-means clustering algorithm is a semi-supervised learning technique.
- ☒ 4. Decision tree algorithm is a semi-supervised learning technique.

Question ID : 630680979068
Option 1 ID : 6306803840249
Option 2 ID : 6306803840251
Option 3 ID : 6306803840248
Option 4 ID : 6306803840250
Status : Not Answered
Chosen Option : --

Q.66 The Bernoulli equation is derived from the principle of:

- Ans ☒ 1. conservation of angular momentum
- ☒ 2. conservation of mass
- ☒ 3. conservation of energy
- ☒ 4. conservation of momentum

Question ID : 630680972490
Option 1 ID : 6306803813742
Option 2 ID : 6306803813739
Option 3 ID : 6306803813741
Option 4 ID : 6306803813740
Status : Answered
Chosen Option : 3

Q.67 In a chemical process, the purpose of a purge stream is to:

- Ans ☒ 1. remove inert substances
- ☒ 2. recycle unused feedstock
- ☒ 3. ensure stable temperature
- ☒ 4. accelerate the reaction rate

Question ID : 630680970612
Option 1 ID : 6306803806708
Option 2 ID : 6306803806709
Option 3 ID : 6306803806710
Option 4 ID : 6306803806707
Status : Answered
Chosen Option : 1

Q.68 Vinyl chloride is produced by adding chlorine to ethylene in the presence of which catalyst?

- Ans
- ☐ 1. Magnesium chloride
 - ☐ 2. Zinc chloride
 - ☐ 3. Silver chloride
 - ☒ 4. Iron chloride

Question ID : 630680959093
Option 1 ID : 6306803760169
Option 2 ID : 6306803760171
Option 3 ID : 6306803760170
Option 4 ID : 6306803760168
Status : Answered
Chosen Option : 3

Q.69 The 'six-tenth-factor' rule for cost estimation is a plot of:

- Ans
- ☒ 1. equipment cost versus capacity on a log-log scale
 - ☐ 2. operating cost versus equipment cost on a linear scale
 - ☐ 3. equipment efficiency versus capacity on a logarithmic scale
 - ☐ 4. total project cost versus time on a semi-log scale

Question ID : 630680959322
Option 1 ID : 6306803761072
Option 2 ID : 6306803761073
Option 3 ID : 6306803761075
Option 4 ID : 6306803761074
Status : Answered
Chosen Option : 1

Q.70 Which type of heat exchanger design typically achieves the optimum heat transfer surface area?

- Ans
- ☐ 1. Parallel flow shell and tube heat exchanger
 - ☐ 2. Counterflow double pipe heat exchanger
 - ☒ 3. Counterflow shell and tube heat exchanger
 - ☐ 4. Parallel flow double pipe heat exchanger

Question ID : 630680959268
Option 1 ID : 6306803760863
Option 2 ID : 6306803760860
Option 3 ID : 6306803760862
Option 4 ID : 6306803760861
Status : Answered
Chosen Option : 3

Q.71 In mass transfer between two fluids in intimate contact, what assumption is commonly made about the interface?

- Ans
- ☐ 1. There is no mass transfer across the interface.
 - ☒ 2. There is equilibrium across the interface.
 - ☐ 3. The interface contains significant mass.
 - ☐ 4. The interface creates a thermal barrier.

Question ID : 630680959901
Option 1 ID : 6306803763524
Option 2 ID : 6306803763521
Option 3 ID : 6306803763523
Option 4 ID : 6306803763522
Status : Answered
Chosen Option : 1

Q.72 Which approach to carbon accounting focuses on monitoring direct emissions from a territory?

- Ans
- ☐ 1. Voluntary-based approach
 - ☐ 2. Carbon footprint approach
 - ☐ 3. Consumption-based approach
 - ☒ 4. Territory-based approach

Question ID : 630680959871
Option 1 ID : 6306803763416
Option 2 ID : 6306803763413
Option 3 ID : 6306803763414
Option 4 ID : 6306803763415
Status : Answered
Chosen Option : 2

Q.73 In optimisation simulators, what is the purpose of including design specifications during iterative calculations?

- Ans ☒ 1. To increase the number of recycle iterations
- ☒ 2. To guide the optimisation process and achieve convergence
- ☒ 3. To replace the optimisation algorithm
- ☒ 4. To simplify the optimisation algorithm

Question ID : 630680959892
Option 1 ID : 6306803763491
Option 2 ID : 6306803763489
Option 3 ID : 6306803763490
Option 4 ID : 6306803763492
Status : Answered
Chosen Option : 2

Q.74 In the context of heat transfer, conduction is the process by which heat is transferred:

- Ans ☒ 1. by mixing of hot and cold fluids
- ☒ 2. through direct molecular interaction
- ☒ 3. by electromagnetic waves
- ☒ 4. through the movement of fluids

Question ID : 630680974074
Option 1 ID : 6306803819934
Option 2 ID : 6306803819933
Option 3 ID : 6306803819932
Option 4 ID : 6306803819931
Status : Answered
Chosen Option : 2

Q.75 The rate of return on investment is calculated by:

- Ans ☒ 1. multiplying yearly profit by total initial investment
- ☒ 2. adding total initial investment to yearly profit
- ☒ 3. subtracting total initial investment from yearly profit
- ☒ 4. dividing yearly profit by total initial investment

Question ID : 630680959341
Option 1 ID : 6306803761149
Option 2 ID : 6306803761151
Option 3 ID : 6306803761150
Option 4 ID : 6306803761148
Status : Answered
Chosen Option : 4

Q.76 What is the role of process flow diagrams in chemical process simulation?

- Ans ☒ 1. They serve as decorative illustrations of chemical processes.
- ☒ 2. They provide a visual representation of equipment without any functional purpose.
- ☒ 3. They summarise connections between equipment and operations.
- ☒ 4. They are used to showcase the artistic design of chemical plants.

Question ID : 630680959881
Option 1 ID : 6306803763452
Option 2 ID : 6306803763451
Option 3 ID : 6306803763449
Option 4 ID : 6306803763450
Status : Answered
Chosen Option : 3

Q.77 Which of the following statements is true about unsupervised learning techniques?

- Ans ☒ 1. Principal component analysis (PCA) is a supervised learning algorithm.
- ☒ 2. Principal component analysis (PCA) is used for increasing the number of features of the data set.
- ☒ 3. Number of clusters must be specified in advance for the process of hierarchical clustering.
- ☒ 4. Number of clusters must be specified in advance for the process of K-means clustering.

Question ID : 630680978890
Option 1 ID : 6306803839546
Option 2 ID : 6306803839548
Option 3 ID : 6306803839545
Option 4 ID : 6306803839547
Status : Not Answered
Chosen Option : --

Q.78 Which of the following is a property of a pure substance?

- Ans ☒ 1. Mixture composition
- ☒ 2. Specific volume
- ☒ 3. Concentration of solutes
- ☒ 4. Boiling point of a solution

Question ID : 630680971373
Option 1 ID : 6306803809591
Option 2 ID : 6306803809592
Option 3 ID : 6306803809594
Option 4 ID : 6306803809593
Status : Answered
Chosen Option : 1

Q.79 Which of the following best describes the Second Law of Thermodynamics?

- Ans ☒ 1. Energy can neither be created nor destroyed.
- ☒ 2. In any energy conversion, some energy is lost as unusable heat.
- ☒ 3. For any spontaneous process, the entropy of the universe decreases.
- ☒ 4. The total energy of an isolated system remains constant.

Question ID : 630680970928
Option 1 ID : 6306803807963
Option 2 ID : 6306803807964
Option 3 ID : 6306803807966
Option 4 ID : 6306803807965
Status : Answered
Chosen Option : 4

Q.80 Which of the following statements regarding the role of momentum in mass transfer is/are correct?

Statement A: In mass transfer, momentum transfer between fluid layers is primarily described by the concept of viscosity.

Statement B: The Reynolds number is a dimensionless quantity that characterises the relative importance of inertial forces to viscous forces in a fluid flow.

- Ans ☒ 1. Only Statement B
- ☒ 2. Both Statement A and Statement B
- ☒ 3. Only Statement A
- ☒ 4. Neither Statement A nor Statement B

Question ID : 630680974708
Option 1 ID : 6306803822609
Option 2 ID : 6306803822610
Option 3 ID : 6306803822608
Option 4 ID : 6306803822611
Status : Answered
Chosen Option : 2

Q.81 Radiation heat transfer occurs through:

- Ans ☒ 1. phase change
- ☒ 2. direct contact between materials
- ☒ 3. electromagnetic waves
- ☒ 4. fluid motion

Question ID : 630680974158
Option 1 ID : 6306803820290
Option 2 ID : 6306803820287
Option 3 ID : 6306803820289
Option 4 ID : 6306803820288
Status : Answered
Chosen Option : 3

Q.82 Which of the following statements about residence time distribution (RTD) in a reactor is NOT true?

- Ans ☒ 1. RTD is useful for identifying the presence of dead zones in a reactor.
- ☒ 2. RTD helps in analysing deviations from ideal reactor behaviour.
- ☒ 3. RTD can provide insights into the flow patterns within a reactor.
- ☒ 4. RTD can be used to predict the exact conversion rate of a reaction.

Question ID : 630680983936
Option 1 ID : 6306803859790
Option 2 ID : 6306803859792
Option 3 ID : 6306803859789
Option 4 ID : 6306803859791
Status : Answered
Chosen Option : 4

Q.83 In a binary mixture, the diffusion coefficient of component A in B is affected by:

- Ans ☒ 1. concentration of component B
☒ 2. only pressure
☒ 3. temperature and pressure
☒ 4. only temperature

Question ID : 630680974630
Option 1 ID : 6306803822259
Option 2 ID : 6306803822258
Option 3 ID : 6306803822256
Option 4 ID : 6306803822257
Status : Answered
Chosen Option : 3

Q.84 What is the basis for solving problems 'by analogy' in transport phenomena?

- Ans ☒ 1. Complex mathematical formulations
☒ 2. Experimental data
☒ 3. Independence of governing processes
☒ 4. Similarity of the governing equation

Question ID : 630680959900
Option 1 ID : 6306803763518
Option 2 ID : 6306803763519
Option 3 ID : 6306803763520
Option 4 ID : 6306803763517
Status : Answered
Chosen Option : 4

Q.85 In a feedforward control system, what is the role of the feedforward transfer function?

- Ans ☒ 1. It models the relationship between the setpoint and the process variable.
☒ 2. It models the disturbance input to the system.
☒ 3. It models the relationship between the disturbance and the process variable.
☒ 4. It models the relationship between the controller output and the process variable.

Question ID : 630680952600
Option 1 ID : 6306803735029
Option 2 ID : 6306803735032
Option 3 ID : 6306803735030
Option 4 ID : 6306803735031
Status : Not Answered
Chosen Option : --